

INFLUENCE OF GRAVITY WAVE ACTIVITY ON LOWER THERMOSPHERIC PHOTOCHEMISTRY AND COMPOSITION

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Abstract—The wind and temperature oscillations of internal gravity waves can cause horizontal variations of a factor of two in minor gas number densities in the lower thermosphere over length scales of several hundred kilometers. The variations are due both to vertical transport of constituents whose lifetimes are long compared to the wave period and to chemical activity driven by temperature dependent reaction rate coefficients. The nightglow emission of the hydroxyl radical provides a remote sensor of wave activity between 80 and 90 km. Theoretical calculations show that the horizontal variations in the atomic hydrogen distribution are the largest single contributor to wave structure in the nightglow followed by the effects of temperature fluctuations on the rate coefficient of the reaction $\text{H} + \text{O}_3 \rightarrow \text{O}_2 + \text{OH}(\nu' > 0)$.

INTRODUCTION

The presence of small scale motions in the lower thermosphere which vary with periods much less than one day has been known since the meteor trail studies of Liller and Whipple (1954). Hines (1960) first interpreted these winds as internal gravity waves and since this time numerous observations of the motions themselves and airglow emissions from the lower thermosphere appear to confirm the concept (Manson and Meek, 1976; Moreels and Herse, 1977; Krassovsky and Shagaev, 1977). Rotational temperatures deduced by Noxon (1978) from emissions of OH and $\text{O}_2(^1\Sigma_g^-)$ indicate that near 95 km variations as large as ± 50 K about a mean value occur on occasion and have a period of 1–2 h. The altitude dependence of the wind amplitudes is as expected from linear gravity wave theory, being $e^{z/2H}$ where z is altitude and H is the atmospheric scale height. Near 115–125 km viscous damping becomes significant and wave energy is dissipated as heat (Hines, 1965).

Periodic vertical velocities cause non-negligible changes in the bulk density of the atmosphere as well as in the number density of minor gases whose time scales for chemical removal are long compared to the wave period. In addition, the large observed temperature variations must have a significant impact on the local photochemistry via associated changes in rate coefficients. The theory of the influence of gravity waves on lower thermospheric airglow emissions has been developed in detail by Weinstock (1978) who then applied the formalism to observed fluctuations in the $\text{O}_2(^1\Sigma)$ nightglow and temperature. The present paper addresses the influence of gravity wave activity on upper atmos-

pheric oxygen–hydrogen chemistry and the OH nightglow using techniques similar to those of Weinstock (1978). Of particular interest is the degree of horizontal structure which exists in minor gas distributions as a result of response to density and temperature fluctuations. Aside from its intrinsic interest such information is required in order to assess the significance of any differences which appear when comparing theoretical predictions with measurements made at a single point in the atmosphere for a brief time period.

THEORETICAL DEVELOPMENT

The forcing functions for the minor constituent variations of interest here are the horizontal and vertical winds and variations of temperature and density characteristic of internal gravity waves. Hines (1960) deduced the properties of such waves from the linearized equations of momentum and mass conservation and the equation of state. Principal assumptions are: (a) the atmosphere is isothermal and (b) there are no background winds. The observation of motions which closely resemble those predicted theoretically indicates that the violation of these assumptions by the true atmosphere does not radically alter the character of the waves. Table 1 presents the equations which govern the oscillations taken or derived from Hines (1960) except that all complex variables have been replaced by their real parts. The “reduced amplitude” A in cm s^{-2} must be determined from observation. The quantity A is referred to as the “reduced amplitude” to distinguish it from the altitude dependent amplitude which, from Table 1, increases as $\exp(z/2H)$. The temperature amplitude

TABLE 1. RELATIONS WHICH GOVERN BULK ATMOSPHERE BEHAVIOR IN THE PRESENCE OF INTERNAL GRAVITY WAVES

Property	Equation
Bulk density perturbation	$M' = \bar{M}(z) A e^{z/2H} \{ \omega^2 k_z \cos \theta + [(1 - \gamma/2) k_y^2 g + (\omega^2 - k_y^2 C^2) 1/2H] \sin \theta \}$
Horizontal velocity	$v = A e^{z/2H} \omega k_y C^2 \{ k_z \cos \theta + [(1 - \gamma/2) g/C^2] \sin \theta \}$
Vertical velocity	$w = A e^{z/2H} \omega (\omega^2 - k_y^2 C^2) \cos \theta$
Temperature perturbation	$T' = \bar{T}(z) A e^{z/2H} (\gamma - 1) \times \{ \omega^2 k_z \cos \theta - g [\gamma \omega^2/C^2 - k_y^2] \sin \theta \}$
Dispersion relation	$\omega^4 - \omega^2 C^2 (k_y^2 + k_z^2) + (\gamma - 1) g^2 k_y^2 - \gamma^2 g^2 \omega^2 / 4C^2 = 0$

ω = wave frequency; k_y = horizontal wavenumber; k_z = vertical wavenumber; $\theta = \omega t - k_y y - k_z z$; y = horizontal distance; z = altitude; g = acceleration due to gravity, C = speed of sound; $\gamma = c_p/c_v$ ratio of specific heats; M , T = mean state bulk density and temperature; H = bulk atmosphere scale height; A = constant which determines wave amplitude.

of ± 50 K at 95 km with a vertical wavelength of 20 km observed by Noxon (1978) implies a value $A = 7\text{--}8 \times 10^6 \text{ cm}^2 \text{ s}^{-2}$ in Table 1 and a horizontal wavelength of 369–370 km. Vertical shear in the mean background wind and temperature gradients lead to downward reflection of select wave-modes before their amplitudes grow sufficiently to cause significant effect. The data of Noxon (1978), however, provide proof that the modes associated with a period of one to two hours ($\omega = 1.16 \times 10^{-3} \text{ s}^{-1}$ for a 1.5 h period) and horizontal and vertical wavenumbers of $k_y \approx 1.70 \times 10^{-7} \text{ cm}^{-1}$ and $k_z \approx 3.14 \times 10^{-6} \text{ cm}^{-1}$ respectively reach the lower thermosphere, at least under certain conditions. The present study is restricted to this single mode and adopts the value $A = 4 \times 10^6 \text{ cm}^2 \text{ s}^{-2}$, which characterizes a large amplitude wave but is still a factor of two below the maximum observed by Noxon (1978).

Figures 1 and 2 illustrate the variability expected in the 70–110 km temperatures due to a wave with $A = 4.0 \times 10^6 \text{ cm}^2 \text{ s}^{-2}$. The mean state temperature, shown as the dotted line in Fig. 1, is adopted from the *US Standard Atmosphere* 1976. Figure 1 presents temperature profiles at the same instant of time but separated by 200 km in the horizontal. Below 80 km the difference between the instantaneous and mean temperatures is less than 7 K. In the lower thermosphere the wave amplitude grows rapidly to $\pm 20\text{--}25$ K at 90 km and ± 40 K at 100 km. Figure 2 illustrates the horizontal temperature structure encountered at a fixed time for several altitudes. At 80 km the temperature varies by 10–15 K over a horizontal distance of 190–200 km.

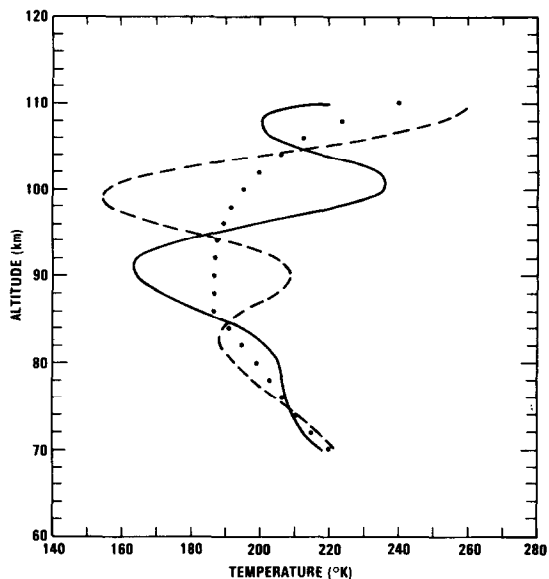


FIG. 1. VERTICAL PROFILES OF TEMPERATURE FOR THE MEAN ATMOSPHERIC STATE (DOTTED LINE) AND IN THE PRESENCE OF AN INTERNAL GRAVITY WAVE WITH $A = 4 \times 10^6 \text{ cm}^2 \text{ s}^{-2}$.

The solid and dashed lines give profiles separated by 200 km in the horizontal.

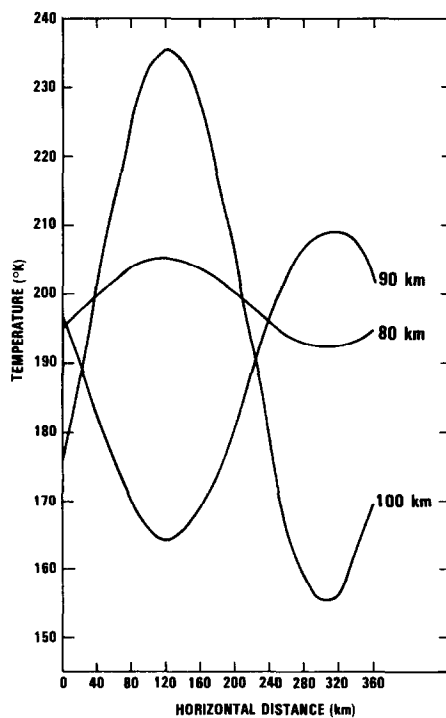


FIG. 2. VARIATION OF TEMPERATURE IN THE HORIZONTAL AT 80, 90, AND 100 km ALTITUDE IN THE PRESENCE OF A GRAVITY WAVE.

This difference increases to 80 K at 100 km. The extremes of vertical velocity at 100 km are $\pm 2.5 \text{ ms}^{-1}$ and at 80 km are $\pm 0.8 \text{ ms}^{-1}$. Such large values upset the instantaneous hydrostatic balance and are accompanied by the bulk density variations presented in Fig. 3. The full range in density encountered at 80 km is 7% of the mean value and increases to 43% at 100 km.

The influence of wave activity on local photochemistry must now be considered. Table 2 presents the photochemical reactions of an oxygen-hydrogen atmosphere included in this study where all rate coefficients are from Hampson and Garvin (1978). It is convenient to consider two families of constituents, odd oxygen, $\text{O}_x = \text{O} + \text{O}_3$, and odd hydrogen, $\text{HO}_x = \text{H} + \text{OH} + \text{HO}_2$. The time scales associated with interconversions of the members of a given family are shorter than the gravity wave period considered here, 1.5 h, so that the ratios of the various odd hydrogens and odd oxygens at any instant are given very nearly by the photochemical

TABLE 2. REACTIONS AND RATE COEFFICIENTS

Reaction	Rate coefficient*
$\text{O} + \text{O}_2 + \text{M} \rightarrow \text{O}_3 + \text{M}$	$k_1 = 1.07 \times 10^{-34} \exp(510/T)$
$\text{O}_3 + h\nu \rightarrow \text{O}_2 + \text{O}$	$J_2 = 9.2 \times 10^{-3} \text{ s}^{-1}$
$\text{H} + \text{O}_3 \rightarrow \text{OH} + \text{O}_2$	$k_3 = 1.0 \times 10^{-10} \exp(-516/T)$
$\text{OH} + \text{O} \rightarrow \text{H} + \text{O}_2$	$k_4 = 4.2 \times 10^{-11}$
$\text{H} + \text{O}_2 + \text{M} \rightarrow \text{HO}_2 + \text{M}$	$k_5 = 2.08 \times 10^{-32} \exp(290/T)$
$\text{OH} + \text{O}_3 \rightarrow \text{HO}_2 + \text{O}_2$	$k_6 = 1.5 \times 10^{-12} \exp(-1000/T)$
$\text{HO}_2 + \text{O} \rightarrow \text{OH} + \text{O}_2$	$k_7 = 3.5 \times 10^{-11}$

* Two body rate coefficients have units $\text{cm}^3 \text{ s}^{-1}$. The three body coefficient is in $\text{cm}^6 \text{ s}^{-1}$. All values are from Hampson and Garvin (1978). T = temperature in degrees Kelvin.

equilibrium values

$$\frac{[\text{O}_3]}{[\text{O}]} = \chi = \frac{k_1[\text{O}_2][\text{M}]}{J_2}, \quad (1)$$

$$\frac{[\text{OH}]}{[\text{H}]} = \frac{k_3\chi}{k_4} + \frac{k_5[\text{M}][\text{O}_2]}{k_4[\text{O}]}, \quad (2)$$

$$\frac{[\text{HO}_2]}{[\text{H}]} = \frac{k_3k_6\chi^2}{k_4k_7} + \left(1 + \frac{k_6\chi}{k_4}\right) \frac{k_5[\text{M}][\text{O}_2]}{k_7[\text{O}]}. \quad (3)$$

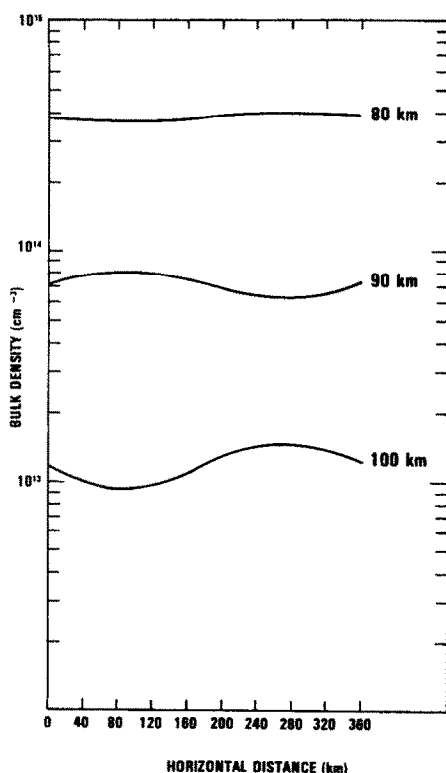


FIG. 3. HORIZONTAL VARIATION IN BULK ATMOSPHERIC DENSITY AT 80, 90 AND 100 km IN THE PRESENCE OF A GRAVITY WAVE.

At the altitudes of interest (70–110 km) atomic oxygen is much more abundant than ozone. The concentration of ozone adjusts to changes in atomic oxygen with a time constant $\tau \sim 1/J_2 = 1.1 \times 10^2 \text{ s}$ which is a factor of 49 less than the wave period. Similar arguments apply to the odd hydrogen family which is predominantly H above 70 km. Fluctuations in the gas ratios occur because of oscillations in $[\text{M}]$, $[\text{O}_2]$, and especially temperature.

In the uppermost mesosphere and above the assumption of a photochemical steady state for total O_x and HO_x breaks down (Park and London, 1974) as the time scales for chemical removal become long compared to the wave period. Under such conditions the transport of total O_x and HO_x by wave motions dominates changes in the chemical production and loss rates due to temperature and density fluctuations. The continuity equation for the number density x , either O_x or HO_x , is

$$\frac{\partial x}{\partial t} + \nabla \cdot \phi_T(x) = P(x) - L(x), \quad (4)$$

where $\phi_T(x)$ is the flux vector representing the effect of motions on all space and time scales, and $P(x)$ and $L(x)$ are the chemical production and loss rates respectively. If one considers x to consist of a mean value which depends only on altitude, $\bar{x}(z)$, plus a perturbation due to wave motions which varies in time, the vertical, and one direction in the horizontal, $x'(y, z, t)$, the following relation is easily

derived:

$$x'(y, z, t) = \bar{x}(z) A e^{z/2H} \left\{ \omega^2 k_z \cos \theta + \left[\left(1 - \frac{\gamma}{2} \right) k_y^2 g - (\omega^2 - k_y^2 C^2) \left(\frac{1}{\bar{x}} \frac{\partial \bar{x}}{\partial z} + \frac{1}{2H} \right) \right] \sin \theta \right\}, \quad (5)$$

where $\theta = \omega t - k_y y - k_z z$. Details of the derivation appear in the Appendix. This result is identical to that for the bulk density perturbation, M' , in Table 1 except that the term $1/\bar{x} \partial \bar{x} / \partial z + 1/2H$ replaces $-1/2H$. This allows for oscillations in a minor constituent to occur out of phase with those in the bulk atmosphere as was noted by Weinstock (1978).

The theory developed above does not include the photochemical production of O_x and HO_x from O_2 and H_2O respectively. Rather, the concern of this study is the fluctuation in minor species about mean values which are specified externally. The mean state profiles of bulk density, molecular oxygen and atomic oxygen are taken from the *US Standard Atmosphere* 1976. At 86 km the computed O distribution of Park and London (1974) was adjusted slightly to merge smoothly with the standard model

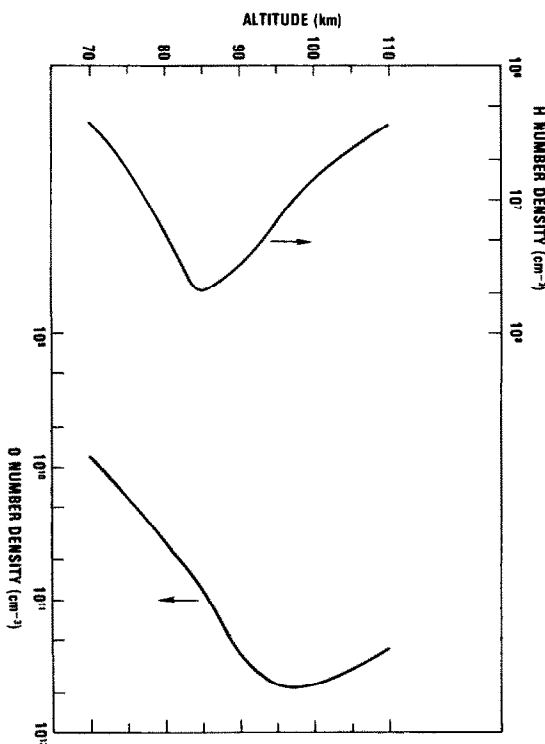


FIG. 4. MEAN STATE PROFILES OF ATOMIC OXYGEN AND ATOMIC HYDROGEN ADOPTED FOR THE CALCULATIONS.

so as to extend the profile down to 70 km. The calculations performed here adopt the atomic hydrogen profile computed by Liu and Donahue (1974). These model results are subject to considerable uncertainty due to the many poorly known inputs required in theoretical calculations. However, such errors are of minor consequence here since emphasis is on fluctuations about the mean state rather than on the processes that determine this state. The adopted mean state O and H profiles appear in Fig. 4.

Figure 5(a) illustrates the horizontal structure induced in the atomic oxygen distribution by a wave of $A = 4 \times 10^6 \text{ cm}^2$ and Fig. 5(b) gives the

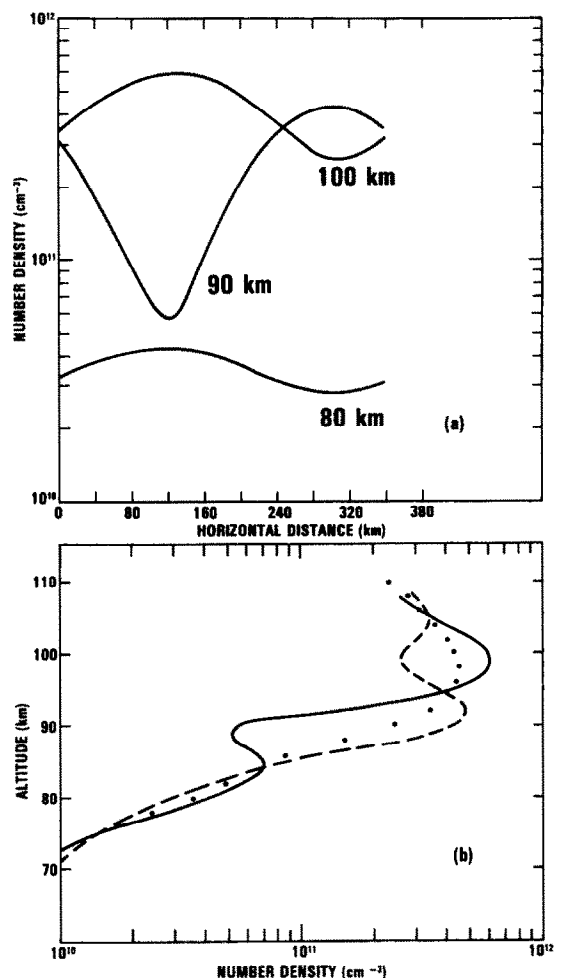


FIG. 5. (a) HORIZONTAL VARIATION IN ATOMIC OXYGEN AT 80, 90, AND 100 km IN THE PRESENCE OF A GRAVITY WAVE. (b) MEAN STATE ATOMIC OXYGEN PROFILE (DOTTED LINE) AND PROFILES SEPARATED BY 200 km IN THE HORIZONTAL UNDER THE INFLUENCE OF WAVE MOTIONS.

vertical profiles separated by 200 km in the horizontal. The magnitude of the oscillation at 90 km is larger than that at 100 km. Such behavior does not conflict with the exponential increase of the gravity wave amplitude with height because a peak in the mean state atomic oxygen occurs near 98 km. Below the maximum the vertical gradient in $[\bar{O}]$ is large, which by equation (5), implies large oscillations. Near 90 km the variations in atomic oxygen, $[O]$, are similar in size to the mean state value, $[\bar{O}]$, so that the linear theory is not strictly applicable. The results of Fig. 5, nonetheless, indicate that minor constituent distributions in the lower thermosphere can exhibit substantial variations over a horizontal scale of order 10^2 km. Near 100 km in altitude the vertical gradient in the mean state atomic oxygen density is small. Thus, despite the large gravity wave amplitude the percent variation in O at 100 km is within a factor of two of that at 80 km.

Figure 6 presents the horizontal distribution of ozone at 80, 90 and 100 km. Equation (1) and the rate coefficient of reaction 1 demonstrate that variations in atomic oxygen and temperature have opposing effects on ozone, with large O promoting large O_3 and high temperature tending to reduce O_3 . At 100 km ozone varies out of phase with both

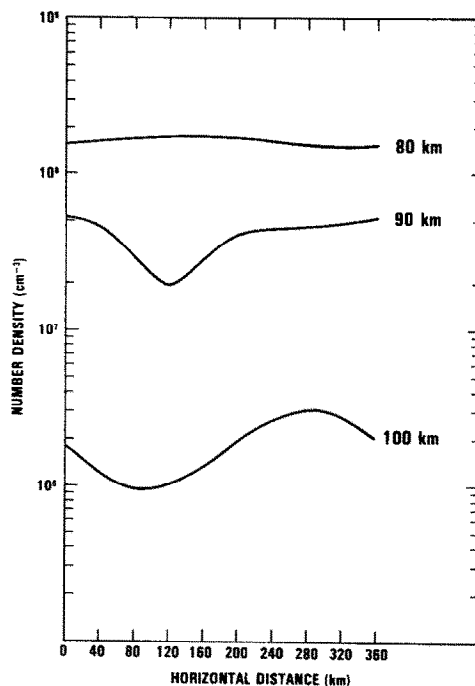


FIG. 6. HORIZONTAL STRUCTURE IN OZONE AT 80, 90, AND 100 km ALTITUDE IN THE PRESENCE OF WAVE MOTION.

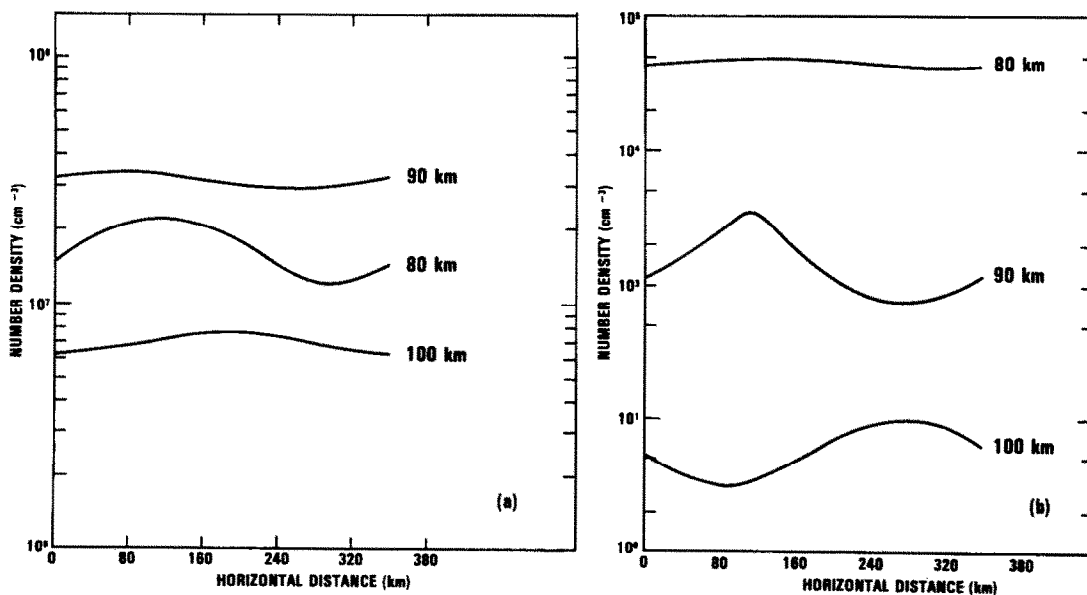


FIG. 7. (a) HORIZONTAL STRUCTURE IN ATOMIC HYDROGEN. (b) HORIZONTAL STRUCTURE IN THE HYDROXYL RADICAL AT 80, 90, AND 100 km.

temperature and atomic oxygen indicating that the temperature dependence of the reaction $O + O_2 + M \rightarrow O_3 + M$ governs the horizontal structure in ozone. At this altitude the temperature fluctuations are sufficiently large to overwhelm the effect due to atomic oxygen. At 90 km the opposite situation prevails. Here the variation in O due to its sharp vertical gradient is a maximum while the temperature variation is much smaller than at 100 km. The sharp minimum in O is pronounced enough to cause similar behavior in O_3 despite the opposing influence of a temperature minimum. At distances removed from the ozone minimum in the horizontal, the temperature and atomic oxygen effects compete so as to destroy a simple sinusoidal ozone variation.

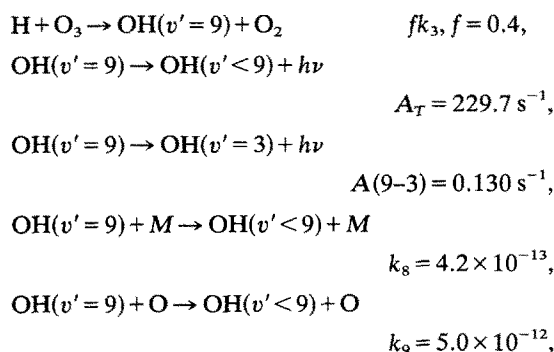
Figure 7(a, b) presents the horizontal structure in the H and OH distributions at 80, 90 and 100 km. The steep vertical gradient in H below the peak in concentration causes large spatial variability at 80 km despite the modest wave amplitude here. Variations at higher altitudes are much less pronounced. The behavior of OH reflects the resultant effects of competing processes. Reference to equation (2) and Table 2 shows that the temperature dependencies of k_3 and χ effectively cancel each other. The second term on the righthand side of equation (5) is the major contributor to OH variability via its inverse dependence on [O]. At 80 km changes in [M] and $[O_2]$ are negligible so that OH variability depends on the ratio $[H]/[O]$ which is nearly constant despite the sizeable amplitudes displayed by the numerator and denominator separately. At 90 km, however, the $[H]$ gradient is small while that in $[O]$ is very large and leads to an OH variation which is opposite that in atomic oxygen. This behavior is further reinforced by the increasing amplitudes in [M] and $[O_2]$, especially at 100 km.

APPLICATION TO THE HYDROXYL RADICAL NIGHTGLOW

The fluctuations in minor gases induced by internal gravity waves are of intrinsic interest; however, it is desirable to link the theoretical results to easily observed atmospheric properties. The Meinel bands of OH which originate between 80 and 90 km in altitude are a well-known feature of the nightglow in which wave structure has been directly observed (Moreels and Herse, 1977; Krassovsky and Shagaev, 1977). Frederick *et al.*, (1978) found variations of a factor of two in the (9-3) band surface brightness observed by satellite at similar local times and attributed these changes to the

action of small scale transport processes. The theoretical interpretation of the observations was based on varying the eddy diffusion coefficient in a one dimensional chemical-transport model while leaving the temperature fixed. However, Frederick *et al.*, (1978) state that a temperature variation of 30-40 K in the 80-90 km region could by itself account for a factor of two change in the (9-3) band volume emission rate near the peak of the layer. The crude treatment of transport provided by the eddy diffusion parameterization does not allow a coupling of temperature and minor gas fluxes. However, the gravity wave theory employed here includes the close coupling between temperature fluctuations and gas fluxes from the outset. The discussion below shows that a gravity wave of reasonably large amplitude ($A = 4 \times 10^6 \text{ cm}^2 \text{ s}^{-2}$) leads to structure in the OH nightglow emission which is readily observable from the ground. The calculations consider only the (9-3) band of the Meinel system; however, the results are trivially generalized to any portion of the OH spectrum.

The reactions which are relevant to the (9-3) band are



where 40% of the reactions of hydrogen atoms with ozone produce OH in the ninth excited vibrational level of the ground electronic term (Streit and Johnston, 1976). Spontaneous transition frequencies, A_T and $A(9-3)$ are as computed by Mies (1974). The quenching rate due to collisions with $M(O_2 \text{ and } N_2)$ is from Streit and Johnston (1976) and that due to atomic oxygen is similar to the value used by Moreels *et al.*, (1977). The total volume emission rate of the 9-3 band, in $\text{cm}^{-3} \text{ s}^{-1}$, is

$$\eta(9-3) = \frac{A(9-3) f k_3 [H][O_3]}{A_T + k_8 [M] + k_9 [O]}. \quad (6)$$

Since ground-based observations must be carried out at night, the O_3 concentration for use in equation (6) cannot be obtained from equation (1).

Rather, after sunset O converts to O₃ according to

$$\frac{\partial[\text{O}_3]}{\partial t} = k_1[M][\text{O}_2][\text{O}] \equiv \frac{[\text{O}]}{\tau}, \quad (7)$$

$$\frac{\partial[\text{O}]}{\partial t} = -\frac{[\text{O}]}{\tau}, \quad (8)$$

which predicts

$$[\text{O}_3(t)] = [\text{O}_3(0)] + [\text{O}(0)](1 - e^{-t/\tau}), \quad (9)$$

where time, t , is measured from sunset. The time constant, τ has the values 2.3×10^4 s, 5.8×10^5 s, and 2.3×10^7 s at 80, 90 and 100 km respectively so that the O to O₃ conversion is essentially completed during the period of darkness at the mesopause, but at 90 km less than 10% of the O is removed. In the altitude region where the OH nightglow originates, 80–90 km, the gravity wave period is much less than the chemical conversion time. The influence of the wave may then be simply included as a rapid variation imposed on the slower ozone increase predicted by equation (9). The daytime atomic hydrogen concentration was assumed to apply for several hours after sunset. The detailed calculations reported by Frederick *et al.*, (1978) show that a slight decrease in the OH(9–3) band volume emission rate occurs in the period after sunset due to the removal of atomic hydrogen, so that the above assumption is not strictly valid. However, since nightglow variability due to gravity wave effects, as opposed to chemical activity, is of primary interest here, a more elaborate treatment is not required.

Figure 8 presents the horizontal structure in several quantities which influence the (9–3) band volume emission rate near its peak, 84–86 km. These include ozone, atomic hydrogen, and the temperature dependent part of k_3 , $e^{-516/T}$. The ozone variation is not sinusoidal due to the competing effects of temperature and atomic oxygen discussed earlier. The bottom panel of Fig. 8 gives the vertical surface brightness in Rayleighs as would be seen by a ground-based observer approximately two hours after sunset. The ratio of maximum to minimum volume emission rate is 2.8. For the temperature dependent term $e^{-516/T}$ this ratio is 1.4 and values for H and O₃ are 1.7 and 1.2 respectively. While the hydrogen and temperature fluctuations are in phase, the ozone contribution tends to shift the maximum volume emission rate slightly in the horizontal. Due to the small ozone amplitude, this effect is not apparent in the nightglow curve. The maximum-to minimum variation of the volume emission rate is slightly less than the product of the

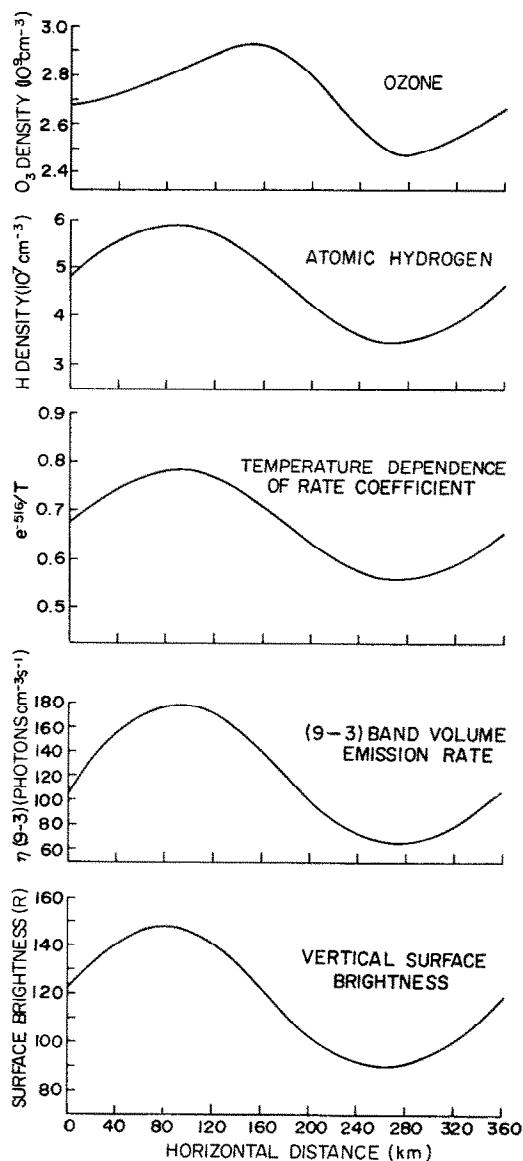


FIG. 8. HORIZONTAL STRUCTURE IN ATMOSPHERIC PROPERTIES WHICH INFLUENCE THE OH NIGHTGLOW EMISSION. All quantities are given at an altitude of 84 km except for the vertically integrated surface brightness.

above ratios because of a small influence of the quenching terms in equation (6) which vary in phase with hydrogen and temperature. The amplitude of the vertical surface brightness is in turn less than that of the volume emission rate due to the slope of lines of constant phase with altitude as indicated in Figs 2 and 3. A partial destructive interference occurs upon vertical integration and

results in a maximum-to-minimum surface brightness ratio of 1.7. Note that the temperature and (9–3) band surface brightness appear to vary in phase in Fig. 8. This result is dependent on the adopted unperturbed profiles of O and H rather than being a general result. The magnitudes of $[O_3]$ and $[H]$ depend on the mean state gradients in $[O]$ and $[H]$ which are model dependent. For example, if the peak of the atomic hydrogen layer shown in Fig. 4 were shifted up or down by 5 km, a phase shift between temperature and the nightglow brightness would be introduced.

CONCLUSIONS

The oscillations associated with large amplitude internal amplitude internal gravity waves above 80 km in altitude can cause variations in minor gas number densities by a factor of two or more over horizontal distances of several hundred kilometers. The response of a given constituent depends on the vertical gradient in its number density and on the extent to which chemistry influences the gas over time scales less than a wave period, in addition to the amplitude of the wave motion. In view of the large variations predicted here, great care must be taken when attempting to validate a theoretical calculation of lower thermospheric composition by comparison to a single measurement.

Nightglow emissions of the hydroxyl radical provide a remote sensor of gravity wave activity between altitudes of 80 and 90 km. Variations in atomic hydrogen induced by vertical wave velocities and changes in the rate coefficient of the reaction $H + O_3 \rightarrow O_2 + OH(\nu')$ due to temperature oscillations are the two largest contributors to wave structure in the nightglow. Horizontal variations in the ozone distribution are somewhat less important. The calculations reported here have assumed only one wave mode to be present in the lower thermosphere. The nightglow observations of Moreels and Herse (1977) show that on occasion this is indeed the case. However, the more common situation is a superposition of many modes of different wavelength, period and amplitude, leading to an apparent chaotic state in the 80 to 100 km altitude region.

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APPENDIX

Mathematical treatment of gravity wave—minor constituent coupling

Weinstock (1978) has given a detailed discussion of the mathematical formalism required to describe the coupling of gravity wave motions with minor constituent and airglow variations. The derivation given below considers only the points which are immediately applicable to the interaction of a single wave mode with an atmospheric gas. The flux vector $\phi_T(x)$ of equation (4) includes all scales of motion while the number density perturbation $x'(y, z, t)$ is defined to represent a response to gravity waves only. If the flux ϕ_T is broken into two portions, one due only to the gravity waves, ϕ_{wave} , and the other, ϕ , due to all other classes of motion,

we have

$$\nabla \cdot \Phi_{\text{wave}} = \frac{\partial}{\partial y}(vx) + \frac{\partial}{\partial z}(wx), \quad (\text{A1})$$

where v and w are as given in Table 1. The continuity equation for the mean state number density, $\bar{x}$, is

$$\nabla \cdot \Phi(\bar{x}) = P(\bar{x}) - L(\bar{x}). \quad (\text{A2})$$

Subtraction of equation (A2) from (4) of the text and use of $x = \bar{x} + x'$ and (A1) produces a continuity equation for x'

$$\begin{aligned} \frac{\partial x'}{\partial t} + \frac{\partial}{\partial y}(v\bar{x}) + \frac{\partial}{\partial y}(vx') + \frac{\partial}{\partial z}(w\bar{x}) + \frac{\partial}{\partial z}(wx') \\ = P(x) - P(\bar{x}) - L(x) + L(\bar{x}). \end{aligned} \quad (\text{A3})$$

By the time constant arguments presented in the text the flux divergence terms in equation (A3) dominate those expressing chemical production and loss when x is taken to be O_x or HO_x . Furthermore, the velocities are perturbation order quantities derived from linear wave theory. It is there-

fore self-consistent, and mathematically convenient, to neglect the products vx' and wx' since number density perturbations induced by such motions should be small compared to the mean state density. A very simple linearized continuity equation results

$$\frac{\partial x'}{\partial t} + \frac{\partial}{\partial y}(v\bar{x}) + \frac{\partial}{\partial z}(w\bar{x}) \approx 0. \quad (\text{A4})$$

The solution of (A4) appears in the text as equation (5). The derivation assumes that the change in number density due to chemical production and loss is negligible compared to bulk transport. The validity of this has been tested by evaluating the perturbation production and loss terms of equation (7) for the reactions in an oxygen-hydrogen atmosphere as given by Liu and Donahue (1974). In all cases, the neglected terms were at least one order of magnitude below the flux divergence. The behavior of total O_x and HO_x in the lower thermosphere over time periods of 1–2 h is therefore well simulated by the continuity equation of a chemically inert gas.